IoT-Based Upload the Measurements of Sensors using ESP8266 on ThingSpeak Cloud

PREPARED BY

Aahan (2023BTech001)

FACULTY SUPERVISOR

Dr. H.P. Agrawal

Department of Computer and Communication Engineering

Institute of Engineering and Technology

JK Lakshmipat University Jaipur

Aim

The aim of this experiment is to design and implement a system using ESP8266 that collects sensor measurements such as temperature, humidity, pressure, and distance, and uploads the data to the ThingSpeak cloud platform for real-time monitoring.

Theory

The ESP8266 NodeMCU is a WiFi-enabled microcontroller used in IoT applications to connect devices to the internet. It enables real-time data communication with cloud platforms. The BME280 sensor measures environmental parameters such as temperature, humidity, and pressure, while the HC-SR04 ultrasonic sensor measures distance using ultrasonic waves. The ThingSpeak platform is used to store and visualize data received from the ESP8266. In this activity, sensor data is collected and transmitted over WiFi to ThingSpeak, where it is displayed in graphical form for analysis.

Working Principle

The ESP8266 first connects to a WiFi network using given credentials. After establishing the connection, it reads temperature, humidity, and pressure values from the BME280 sensor and measures distance using the ultrasonic sensor. The collected data is then sent to the ThingSpeak cloud using HTTP protocol. The cloud platform stores the data and displays it in graphical form, allowing real-time monitoring from anywhere.

Code

```
#include <ESP8266WiFi.h>
#include <Wire.h>
#include <Adafruit_BME280.h>

// WiFi
const char* ssid = "wifi";
const char* password = "alphabet01";

// ThingSpeak
String apiKey = "VEN55UMHIRXD50DK";
const char* server = "api.thingspeak.com";

// BME280
Adafruit_BME280 bme;

// Ultrasonic Pins
#define TRIG D5
#define ECHO D6

WiFiClient client;
```

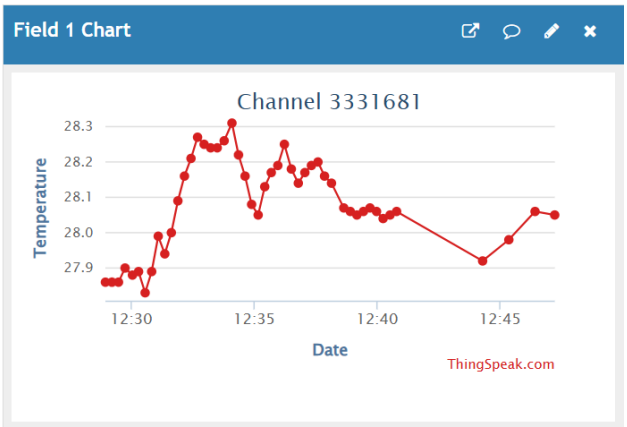
```
void setup() {
  Serial.begin(115200);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  Wire.begin(D2, D1);

  // BME280 init
  if (!bme.begin(0x76)) {
    Serial.println("BME280 not found!");
    while (1);
  }

  // WiFi connect
  WiFi.begin(ssid, password);
  Serial.print("Connecting");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected!");
}

// Ultrasonic function
float getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  long duration = pulseIn(ECHO, HIGH);
  return duration * 0.034 / 2;
}

void loop() {

  // ===== BME280 =====
  float temperature = bme.readTemperature();
  float humidity = bme.readHumidity();
  float pressure = bme.readPressure() / 100.0;
  float altitude = bme.readAltitude(1013.25);
```

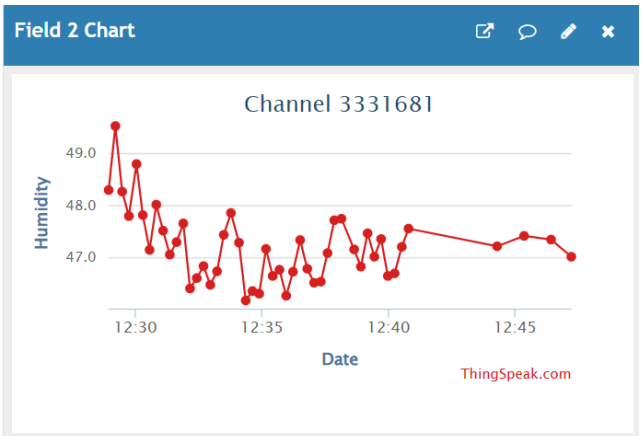
```
  // ===== Ultrasonic =====
  float distance = getDistance();

  Serial.println("---- DATA ----");
  Serial.print("Temp: "); Serial.println(temperature);
  Serial.print("Humidity: "); Serial.println(humidity);
  Serial.print("Pressure: "); Serial.println(pressure);
  Serial.print("Distance: "); Serial.println(distance);

  // ===== ThingSpeak Upload =====
  if (client.connect(server, 80)) {

    String url = "/update?api_key=" + apiKey +
                 "&field1=" + String(temperature) +
                 "&field2=" + String(humidity) +
                 "&field3=" + String(pressure) +
                 "&field4=" + String(altitude) +
                 "&field5=" + String(distance);

    client.print(String("GET ") + url + " HTTP/1.1\r\n" +
                 "Host: " + server + "\r\n" +
                 "Connection: close\r\n\r\n");

    Serial.println("Uploaded!");
  }

  client.stop();

  delay(15000);
}
```

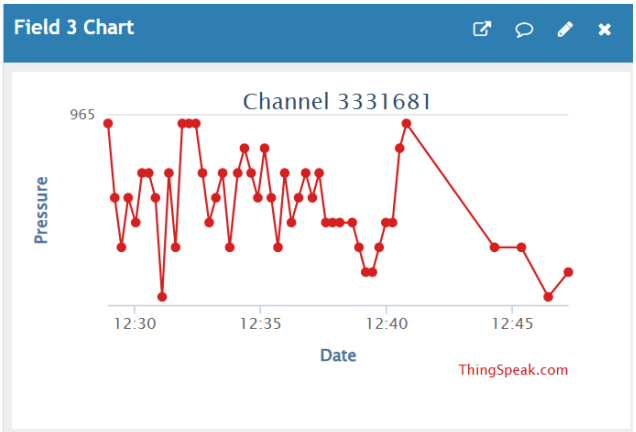
Result

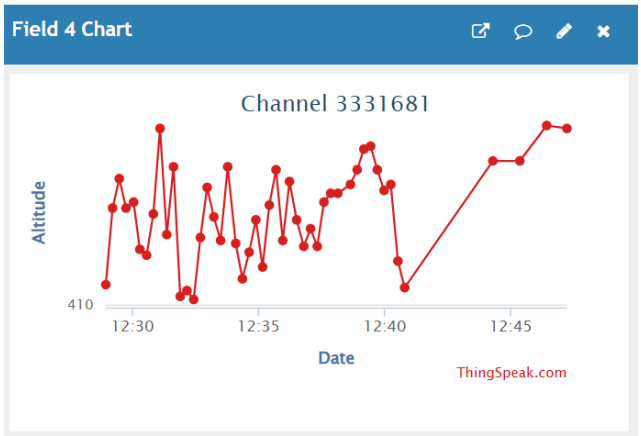
The sensor measurements were successfully collected using ESP8266 and uploaded to the ThingSpeak cloud. The data was displayed in graphical format, enabling real-time monitoring and analysis.

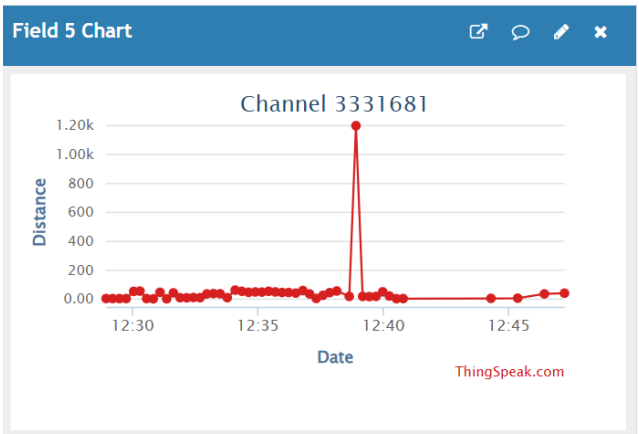
Charts Visualizations











CSV Data

created_at	entry_id	field1	field2	field3	field4	field5
2026-04-08 06:58:57 UTC	1	27.86	48.3	964.96	410.07	4.25
2026-04-08 06:59:13 UTC	2	27.86	49.53	964.93	410.33	3.93
2026-04-08 06:59:29 UTC	3	27.86	48.27	964.91	410.43	3.93
2026-04-08 06:59:45 UTC	4	27.9	47.8	964.93	410.33	3.93
2026-04-08 07:00:03 UTC	5	27.88	48.8	964.92	410.35	53.91
2026-04-08 07:00:18 UTC	6	27.89	47.82	964.94	410.19	55.42
2026-04-08 07:00:34 UTC	7	27.83	47.15	964.94	410.17	3.93
2026-04-08 07:00:50 UTC	8	27.89	48.02	964.93	410.31	2.31
2026-04-08 07:01:06 UTC	9	27.99	47.52	964.89	410.6	47.11
2026-04-08 07:01:22 UTC	10	27.94	47.06	964.94	410.24	2.33
2026-04-08 07:01:38 UTC	11	28	47.3	964.91	410.47	43.32
2026-04-08 07:01:54 UTC	12	28.09	47.66	964.96	410.03	10.17
2026-04-08 07:02:10 UTC	13	28.16	46.41	964.96	410.05	9.49
2026-04-08 07:02:26 UTC	14	28.21	46.61	964.96	410.02	11.22
2026-04-08 07:02:42 UTC	15	28.27	46.84	964.94	410.23	10.17
2026-04-08 07:02:58 UTC	16	28.25	46.48	964.92	410.4	36.07
2026-04-08 07:03:14 UTC	17	28.24	46.74	964.93	410.3	38.37

Conclusion

The activity successfully demonstrates how sensor data can be uploaded to a cloud platform using ESP8266. It highlights the practical implementation of IoT in monitoring environmental conditions and provides a foundation for developing more advanced smart systems.